Project Planning, Project Scheduling, Project Monitoring and Implementation

4.1 Introduction

Planning is answering questions like, what must be done, by whom, for how much, how, when, and so on.

What is a Project Plan?

The project plan decides what, where, who, why, how and when to do the project. The purpose of a project plan is to guide the execution and control the project phases. A project plan is a series of formal documents that define the execution and control stages of a project.

4.2 Nature of Project Planning

The project planning helps in streamlining the process of the Project. Planning helps in the smooth running of the project as every aspect of the project is taken into consideration, and the required solution.

The Project plan consists of three related parts.

1. Scope
2. Schedule
3. Cost

1. Scope: It states the methods and procedures of each work and the name of the person or organisation unit, responsible for the work.

2. Schedule: It states the estimated time required to complete each work and the interrelationships among the work.

3. Cost: It is stated in the project budget, usually called the control budget.

4.3 Need for Project Planning

One of the objectives of project planning is to completely define all the work required so that it will be readily identifiable to each project participant.

There are four basic reasons for project planning.

1. To eliminate or reduce uncertainty.
2. To improve efficiency of the operation.
3. To obtain a better understanding of the objectives.
4. To provide a basis for monitoring and controlling the work.

4.4 Functions of Project Planning

The functions of the Project Planning are;

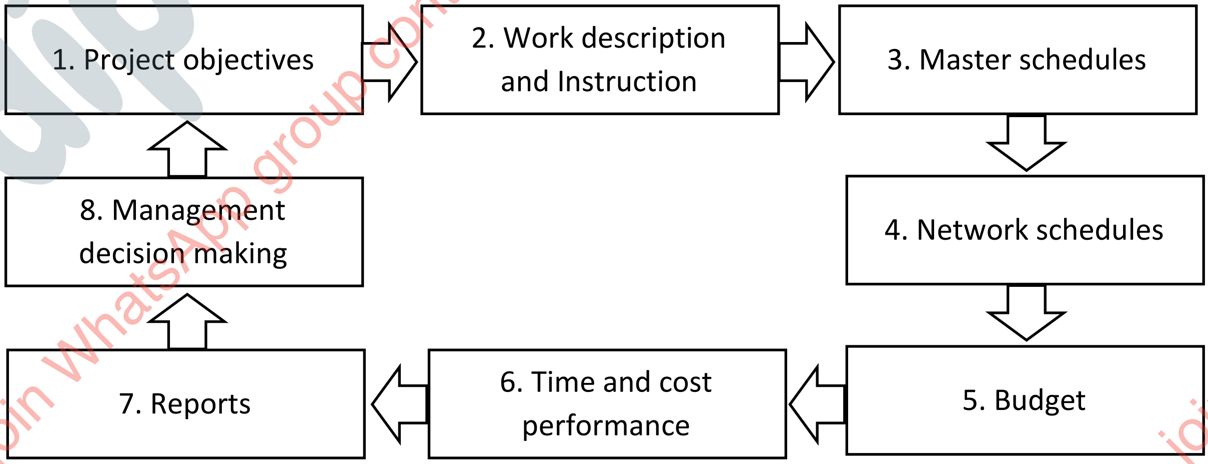
1. It should provide a basis for organising the work on the project and allocating responsibilities to individuals.
2. It is a means of communication and coordination between all those involved in the project.
3. It inspires the people to look ahead.
4. It induces a sense of urgency and time consciousness.
5. It establishes the basis for monitoring and control.

4.5 Steps in Project Planning

1. Define the problem to be solved by the project.
2. Develop a mission statement, followed by statements of major objectives.
3. Develop a project strategy that will meet all project objectives.
4. Write a scope statement to define project boundaries (what will and will not be done).
5. Develop a work breakdown structure (WBS).
6. Using the WBS, estimate activity durations, resource requirements, and costs.
7. Prepare the project master schedule and budget.
8. Decide on the project organization structure.
9. Create the project plan.
10. Get all project stakeholders to sign off on the plan.

4.6 Project Planning Structure

The various activities involved in project planning is given in the following chart. It is called as project planning structure.



Types of Project Plan:

Single use plans: It includes programmes schedules and special ways of operating under particular circumstances. It can also be known as short term plans to deal with the specific problem for specific place with prescribed time limit.

Standing plans: Standing plans are those plans which include policies, standard methods and standard operation, procedures. They are designed to deal with recurring problems. It may be treated as standard document to be used in different plans to deal with a set of problems.

4.7 Project Objectives and Policies

Project objectives are what we plan to achieve by the end of the project. It includes deliverables and assets, increasing productivity or motivation. Project objectives should be attainable, time-bound, measurable specific goals of the project.

An effective project goal has the following characteristics. These characteristics are captured in the term **SMART**, an acronym for the aspects of a goal commitment. These characteristics of a project goals are,

Specific – Clear about what, where, when and how.

Measurable - Are we able to measure the problem, establish a baseline, and set targets for improvement?

Achievable – Is the objectives are attainable?

Realistic – Is the project objectives and schedules are realistic?

Time Bound – Have we set the time for completion?

The objectives of a project may be

1. Technical objectives.
2. Performance objectives.
3. Time and cost objectives.

Project Policies:

Policies are the general guide for decision making on individual actions. Some of the policies of a project are,

1. Extent of work given to outside contractors.
2. Number of contracts to be employed.
3. Terms of the contract etc.

Project policies must be formulated on the basis of following principles:

1. It must be based upon the known principles in the operating areas.
2. It should be complementary for co-ordination.
3. It should be definite, understandable and preferably in writing.
4. It should be flexible and stable.
5. It should be reasonably comprehensive in scope.

4.8 Tools of Project Planning

The tools for project planning may be grouped into two categories. Traditional tools and network analysis tools.

Project Planning tools are,

1. Gantt Chart
2. Network Diagrams
3. Critical Path Methods (CPM)
4. Work Breakdown Structure (WBS)
5. PERT chart

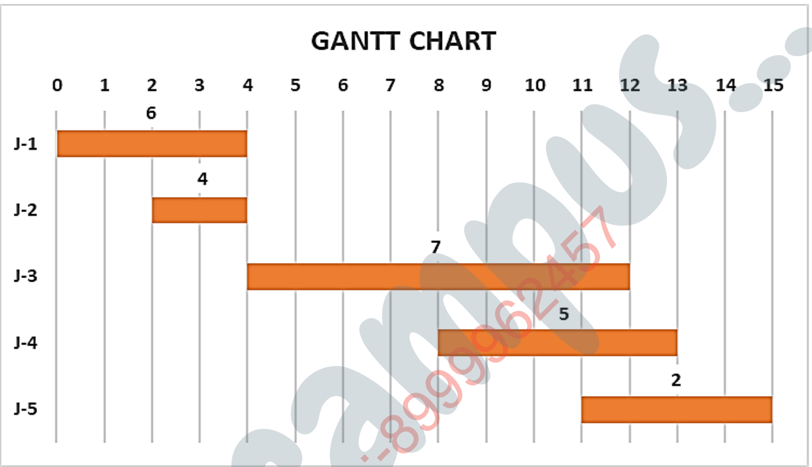
Gantt Chart:

It is the oldest formal planning tool designed by Henry Gantt in 1993. Under this, the activities of project are broken down into a series of well-defined jobs of short duration whose cost and time can be estimated. It is a pictorial device in which the activities, jobs are represented by horizontal bars in the time axis. The length of the bar indicates the estimated time for the job. The left-hand end of the bar shows the beginning time, the right-hand end shows the ending time. The manpower required for the activity is shown by a number on the bar.

An illustrative Gantt chart is shown as follows.

PROJECT DETAILS

JOBS	START DAY	DURATION	MANPOWER
J-1	0	4	6
J-2	2	2	4
J-3	4	8	7
J-4	8	5	5
J-5	11	4	2



The merits and demerits of Gantt chart are below:

MERITS:

1. It is simple to understand.
2. It can be used to show progress.
3. It can be used for manpower planning.

DEMERITS:

1. It cannot show inter-relationship among activities on large, complete projects.
2. There may be physical limit to the size of the bar chart.
3. It cannot easily cope with frequent changes or updating.

Network Diagram:

A network diagram is a graphical representation of all the activities of a project, placing them in their proper sequence and with all interdependencies clearly established. The network diagram provides a complete picture of the project.

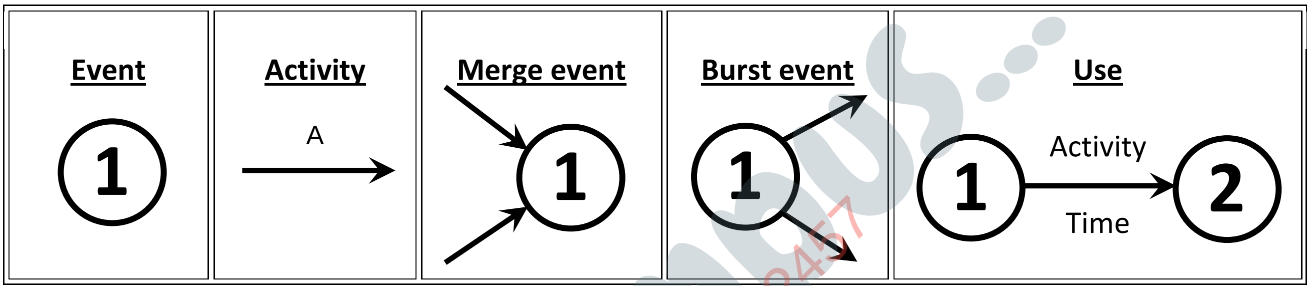
Components of network diagram:

Activities

1. Real or dummy.
2. Predecessor-successor relationship.
3. Represented by arrows.

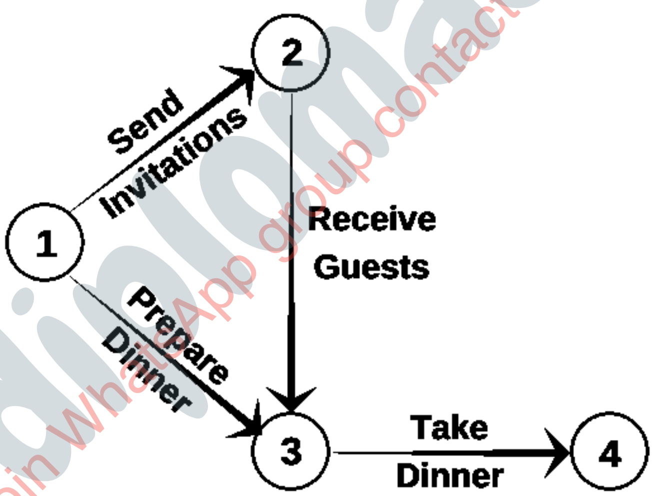
Events

1. Instantaneous occurrence.
2. Denotes the beginning or end of an activity.
3. Represented by circles.
4. Burst or merge events.



In the network techniques, the activities, events and their inter relationships are represented by a network diagram which is also called an arrow diagram.

Example:



EVENT	ACTIVITY
Invitation (1-2)	Sending
Dinner (1-3)	Preparing
Guests' arrival (2-3)	Receiving
Dinner (3-4)	Taking

The advantages of network diagram are:

1. They can effectively handle inter relationships among project activities.
2. They identify the activities which are critical to them.
3. Completion of the project on time indicate the float (spare time) for other activities.
4. They can handle very large and complex projects.

The disadvantages of network diagram are:

1. They are not easily understood by the project personnel.
2. They do not define an operational schedule which tells who does what and when.

4.9 Project Scheduling

It is one of the key components in the project control system. It refers to when it is to be done and how much is to be done. The purpose of scheduling is to obtain commitment, communicate the commitments to concerned project team and ensure coordination among them. The scheduling is helpful to link the summary of activities and review the lapses.

Purpose: The ongoing scheduling and monitoring process enables us to:

1. Successively detail out the schedule to provide physical equivalence with reality.
2. Adopt the schedule to the changed realities.
3. Provide intervention when stability of the work system is being threatened and re-energize the system.

4.10 Time Monitoring Efforts

Monitoring is an action inducing efforts to ensure that, commitments made by various agencies are followed by action for seamless execution.

For monitoring the time aspects of the project, the below efforts should be taken.

1. Development of project execution plan and overall project implementation schedule.
2. Preparation of special condition of contract for scheduling and monitoring.

3. Evaluation of bids in relation to scheduling and monitoring.
4. Review the detailed schedules and progress reports submitted by vendors and contractors.
5. Reviews with owner, consultants, contractors and vendors.
6. Project audit and corporate review.
7. Monthly progress report to the owners.
8. Installation and operation of an on-line information system.
9. On job training for on-going schedule and monitoring.

4.11 Bounding Schedules

Scheduling of non-critical activities can be done by two methods.

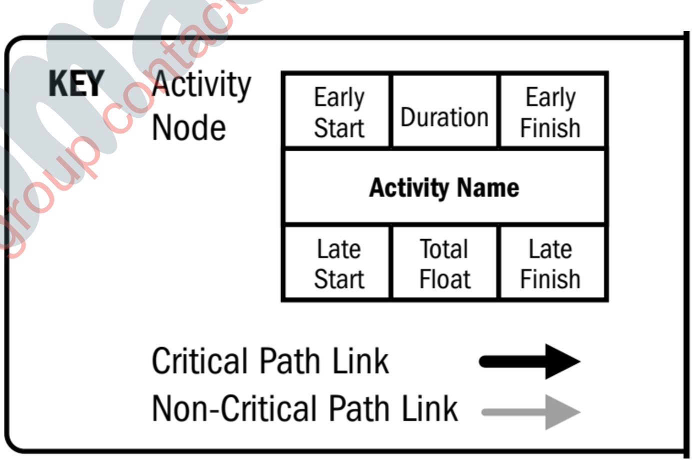
1. Early start schedule
2. Late start schedule

Early start schedule:

The early start schedule indicates a cautious attitude towards the project and a desire to minimise the possibility of delay. It provides a greater measure of protection against uncertainties and adverse circumstances. Early start schedule refers to the schedule in which all activities start as early as possible.

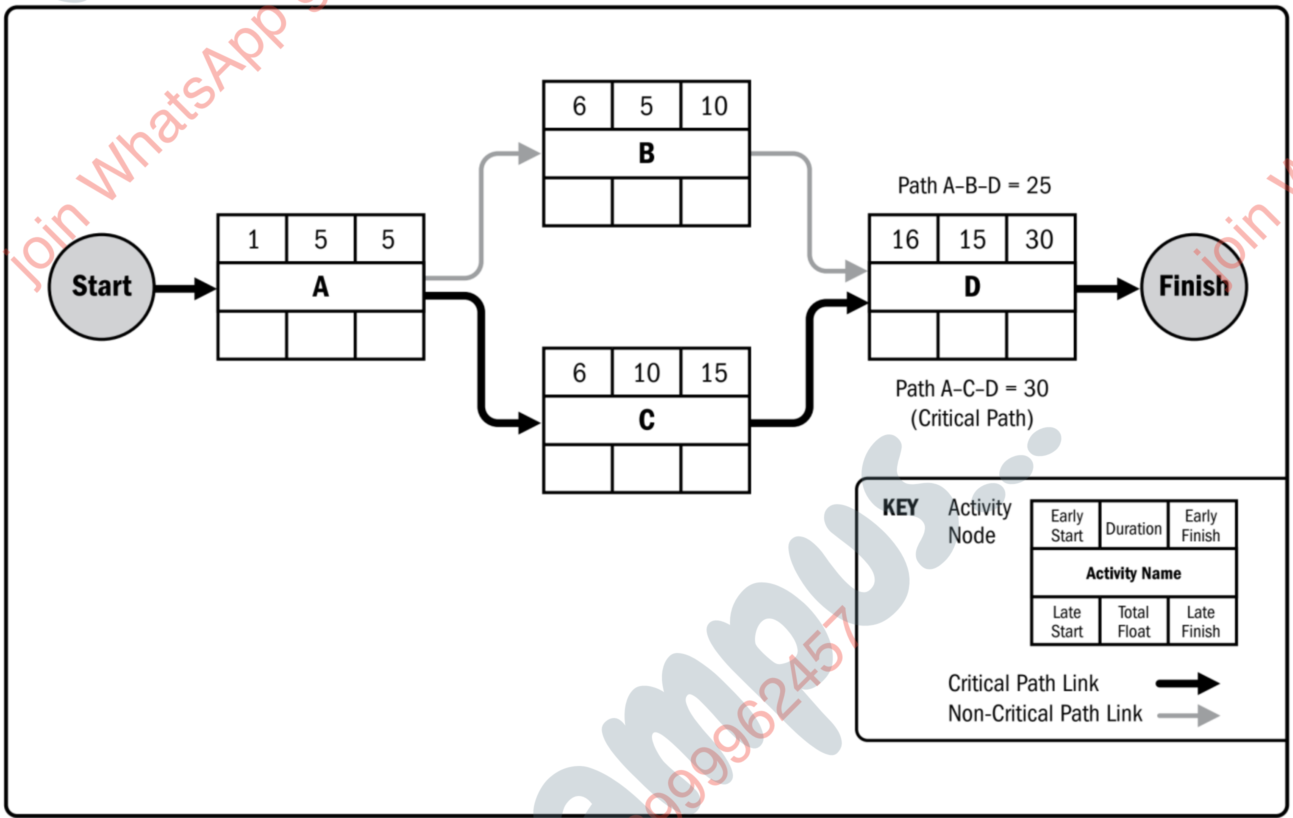
In this schedule,

1. All events occur at their earliest because all activities start at their earliest starting time and finish at their earliest finishing time.
2. There may be time lags between the completion of certain activities.
3. All activities emerging from an event begin at the same time.



A model for early start schedule is given below:

ACTIVITY	PREDECESSOR	DURATION
A	--	5
B	A	5
C	A	10
D	B, C	15



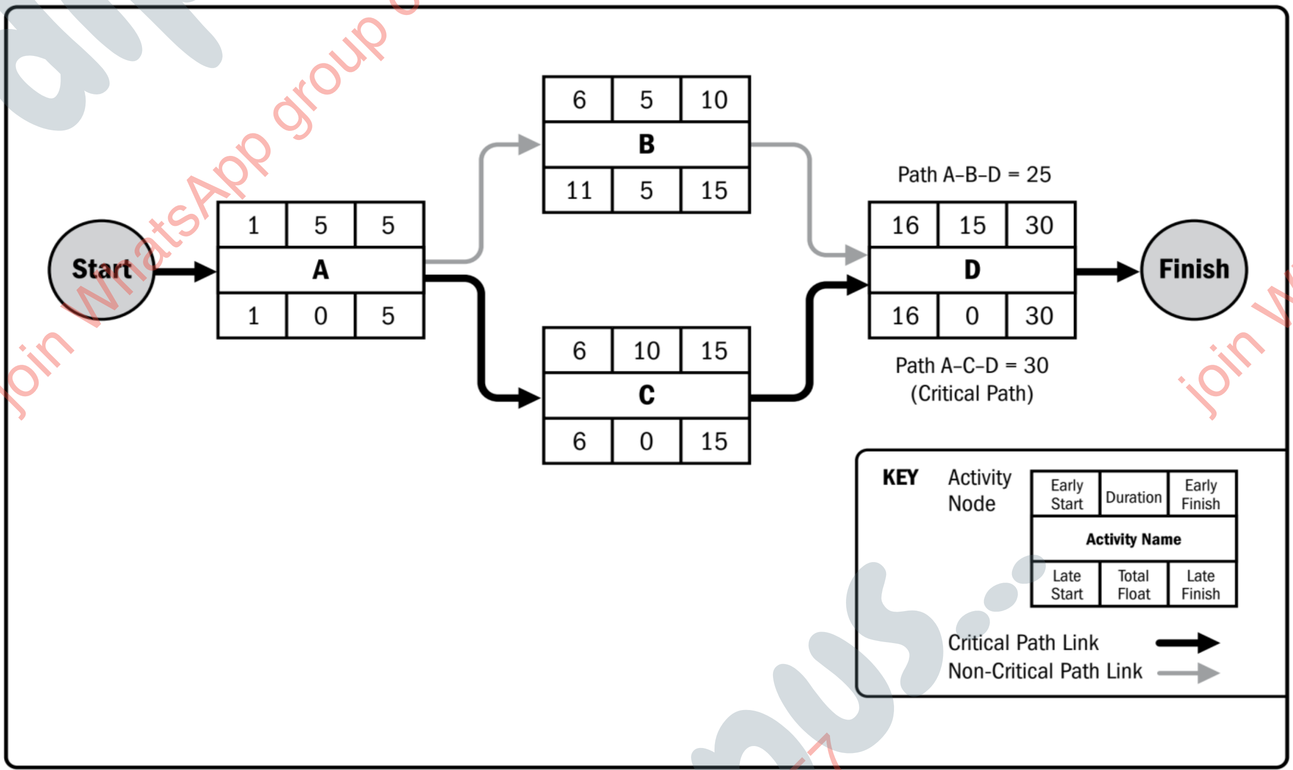
The late start schedule:

It refers to the schedule arrived at when all activities are started as late as possible. In this schedule,

1. All events occur at their latest because all activities start at their latest starting time and finish at their latest finishing time.
2. Some activities may start after a time lag subsequent to the occurrence of the proceeding events.
3. All activities leading to an event are completed at the same time.

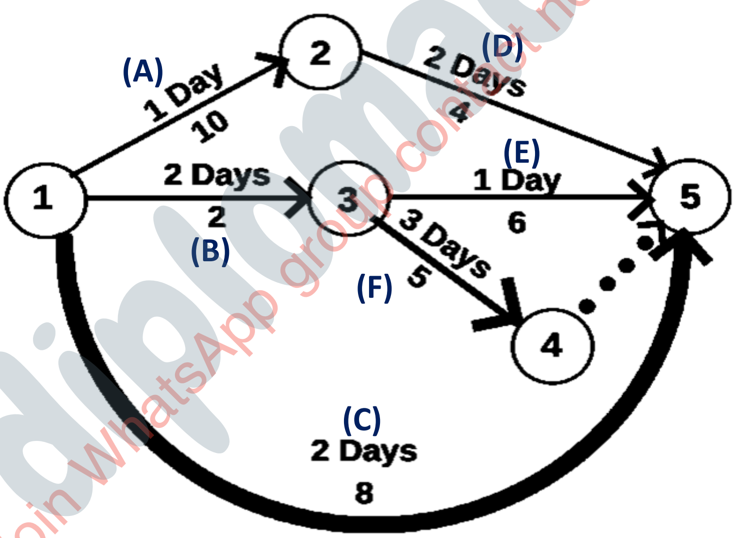
The late start schedule reflects a desire to commit resources as late as possible. However, such a schedule provides no elbow room in the wake of adverse developments. Any unanticipated delay results in increased project duration.

A model for late start schedule is given below:



4.12 Scheduling to Match Availability of Manpower

Let us consider a small project for which the network diagram is shown in fig.



ACTIVITY	DURATION (DAY)	MANPOWER (MEN)
A (1-2)	1	10
B (1-3)	2	2
C (1-5)	2	8
D (2-5)	2	4
E (3-5)	1	6
F (3-4)	3	5

In fig. activity duration is shown above the activity arrow and manpower requirement is shown below the activity arrow. **Only 12 men are available for the project (a manpower resource constraint).**

The early start schedule of this project is shown as a graph on the horizontal time scale in fig.

Early start schedule

DAYS ->	1	2	3	4	5
	A	D	D		
	10	4	4		
	B	B	F	F	F
	2	2	5	5	5
	C	C			
	8	8			
			E		
			6		
TOTAL MEN -->	20	14	15	5	5

Looking at the manpower requirement for the early start schedule we find it is as follows:

20 men for the first day, **14** men for the second day, **15** men for the third day, **6** men for the fourth day and **6** for the fifth day.

Obviously, this schedule is unacceptable in view of the manpower constraint. So, we explore the possibility of shifting activities. Our efforts at shifting activities, keeping the project duration at **five** days soon reveals that no schedule is feasible with only **12** men, so we extend the duration of the project by one day and try various schedules to whether we can find a feasible schedule. A little juggling of activities shows that a schedule like the one shown in fig. is feasible - this is the best we can do.

A feasible schedule

DAYS ->	1	2	3	4	5	6
A	10					
D			4	4		
B	2	2				
F				5	5	5
C		8	8			
E						6
TOTAL MEN -->	12	10	12	9	5	11

4.13 Scheduling to Match Release of Funds

The cost estimates for various activities of a simple project are given in the table.

Cost Estimates:

Activity	Duration (months)	Cost per month	Cost per activity
A (1-2)	13	2,00,000	26,00,000
B (1-3)	12	5,00,000	60,00,000
C (2-4)	2	10,00,000	20,00,000
D (3-4)	8	2,50,000	20,00,000
E (2-5)	15	1,00,000	15,00,000
F (4-5)	2	7,50,000	15,00,000
		Total	**₹ 1,56,00,000**

The government has decided to release **₹ 1,56,00,000** required for the project in the following manner.

₹ 69,00,000 in the first year **₹ 68,00,000** in the second year, and **₹ 19,00,000** in the third year. It has also stipulated that the unspent amount would lapse and hence cannot be carried forward.

Before we develop the project schedule, a preliminary question may be asked: it is possible prima facie to schedule this project without extending its duration beyond 28 months? which is the minimum time required for the given project to complete its activities. To answer this question let us look at the fund's requirement for the early start schedule and late schedule. This is shown in Fig.

Gantt chart: Cumulative funds release scheduling

EARLY START SCHEDULE																																				
	YEAR -1												YEAR-2												YEAR-3											
Months	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
A (1-2)	A (13)																																			
Cost / Month	2	2	2	2	2	2	2	2	2	2	2	2	2																							
B (1-3)	B (12)																																			
Cost / Month	5	5	5	5	5	5	5	5	5	5	5	5																								
C (2-4)														C (2)																						
Cost / Month														10	10																					
D (3-4)													D (8)																							
Cost / Month													2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5																
E (2-5)														E (15)																						
Cost / Month														1	1	1	1	1	1	1	1	1	1	1	1	1	1	1								
F (4-5)																					F (2)															
Cost / Month																					7.5	7.5														
	28 months																																			
Total cost / Month	7	7	7	7	7	7	7	7	7	7	7	7	4.5	14	14	3.5	3.5	3.5	3.5	3.5	8.5	8.5	1	1	1	1	1	1								
Cost / Year	84												68												4											

LATE START SCHEDULE																																				
	YEAR -1												YEAR-2												YEAR-3											
Months	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
A (1-2)	A (13)																																			
Cost / Month	2	2	2	2	2	2	2	2	2	2	2	2	2																							
B (1-3)							B (12)																													
Cost / Month							5	5	5	5	5	5	5	5	5	5	5	5																		
C (2-4)																									C (2)											
Cost / Month																									10	10										
D (3-4)																			D (8)																	
Cost / Month																			2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5										
E (2-5)														E (15)																						
Cost / Month														1	1	1	1	1	1	1	1	1	1	1	1	1	1	1								
F (4-5)																											F (2)									
Cost / Month																											7.5	7.5								
	28 months																																			
Total cost / Month	2	2	2	2	2	2	7	7	7	7	7	7	7	6	6	6	6	6	3.5	3.5	3.5	3.5	3.5	3.5	14	14	8.5	8.5								
Cost / Year	54												58												44											

From the Early start and late start schedules we find that:

1. The rate of expenditure is relatively higher for the earlier stages in the early start schedule and is relatively higher for the later stages in the late start schedule.
2. A rate of spending greater than that of the early start schedule is not possible (This is because in the early start schedule all activities start as early as possible). Any release of funds above the early start — schedule is beyond the capacity of the project to spend.
3. The rate of spending corresponding to the late start schedule is the absolute minimum necessary to complete the project on time. If the rate of spending is less, than that corresponding to the late start schedule the project duration will have to be necessarily extended.
4. A pattern of funds release lying between the two bounds, early start and late start schedule requirements, prima facie suggests that a schedule can be worked out without extending project duration.

Let us now look at the cumulative funds release pattern for our illustrative project. This lies between the early start and late start schedule requirement. So, prima facie it suggests that a feasible schedule without extending the project duration can be developed. Let us proceed further and consider scheduling year by year.

The activities thus begin in year 1 according to the early start schedule are **A (1-2)** and **B (1-3)**. If both these activities are commenced as early as possible, the fund requirement for year I would be 84 lakhs. Since this amount exceeds 69 lakhs, the amount to be released in year 1, the expenditure in year 1 has to be reduced by 15 lakhs. For this we consider the possibility of shifting activities to subsequent periods. Looking at activities **A (1-2)** and **B (1-3)** we find that **A (1-2)** is on the critical path, hence it cannot be shifted. Activity **B (1-3)**, however, can be shifted as it is not on the critical path. Since activity **B (1-3)** requires 5 lakhs per month it has to be shifted by three months so that the amount spent in year 1 is equal to the amount released in that year. Since there is free float of six months for activity **B (1-3)**, we shift it by three months.

In the year 2 the effects of shifting activity **B (1-3)** by three months are as follows:

(i) The funds requirement for year 2 increases by 15 lakhs.

(ii) The earliest starting time for activity **D (3-4)** moves to 15 months from 12 months and the earliest finishing time moves to 23 months from 20 months. Since this shift occurs within year 2, there is no change in funds requirement on account of activity **D (3-4)**.

(iii) The earliest starting time for activity **F (4-5)** moves to 23 months from 20 months. This decreases the funds requirement for year 2 by 7.5 lakhs. Since the funds budgeted for year 2 is only 68 lakhs, we consider the possibility of shifting some activities to year 3. We find that by shifting activity **F (4-5)** to year 3 the expenditure in year 2 can be reduced to 68 lakhs. As a result of this shifting the expenditure for year 3 (first four months of it) equals the budgeted funds release for that year.

Gantt chart: Scheduling to match release of funds

	YEAR -1												YEAR-2												YEAR-3											
Months	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
A (1-2)	A (13)																																			
Cost / Month	2	2	2	2	2	2	2	2	2	2	2	2	2																							
B (1-3)				B (12)																																
Cost / Month				5	5	5	5	5	5	5	5	5	5	5	5																					
C (2-4)														C (2)																						
Cost / Month														10	10																					
D (3-4)																D (8)																				
Cost / Month																2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5													
E (2-5)														E (15)																						
Cost / Month														1	1	1	1	1	1	1	1	1	1	1	1	1	1	1								
F (4-5)																									F (2)											
Cost / Month																									7.5	7.5										
	28 months																																			
Total cost / Month	2	2	2	7	7	7	7	7	7	7	7	7	7	16	16	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	1	8.5	8.5	1	1								
Cost / Year	69												68												19											

4.14 Problems in scheduling Real-Life projects

In real life projects there may be hundreds of activities and there may be several constraints. The problem of scheduling in such cases tends to become very complex. For solving such problems, the technique of linear programming can be used. However, when a problem has numerous activities, the technique of linear programming becomes computationally unwise and expensive, even with the aid of the fastest computers available.

Project Monitoring and Implementation

4.15 Introduction

Monitoring is an integral part of every project, from start to finish.

A project is a series of activities (investments) that aim at solving particular problems within a given time frame and in a particular location. The investments include time, money, human and material resources. Before achieving the objectives, a project goes through several stages. Monitoring should take place at all stages of the project cycle.

The three basic stages of monitoring are:

1. Project planning
2. Project implementation
3. Project evaluation.

Monitoring should be executed by all individuals and institutions which have an interest (stake holders) in the project. To efficiently implement a project, the people who are planning and implementing, should plan for all the interrelated stages.

4.16 Situation Analysis and Problem Definition

Situation analysis is a process through which the general characteristics and problems of the community are identified. It involves the identification and definition of the characteristics and problems specific to particular categories of people in the community. These could be people with disabilities, women, youth, farmer and artisans.

Information necessary to understand the community includes,

1. Population characteristics (e.g., sex, age, tribe, religion and family sizes);
2. Political and administrative structures (e.g., community committees and local councils);
3. Economic activities (including agriculture, trade and fishing);
4. On-going projects related to city, district, Central Government, non-Government organisations (NGO's), and community-based organizations (CBOs);
5. Socio-economic infrastructure (e.g., schools, hospitals, roads etc.);

6. Community organisations (e.g., savings and credit groups, women groups, self-help groups), their functions and activities.

Information for situation analysis and problem definition should be collected with the involvement of the community members using several techniques. This ensures valid, reliable and comprehensive information about the community and its problems.

Some of the following techniques could be used:

1. Document's review;
2. Surveys;
3. Discussions with individuals, specific groups and the community as a whole;
4. Interviews;
5. Observations;
6. Listening to people;
7. Brainstorming;
8. Informal conversations;
9. Making an inventory of community social resources, services and opportunities.

Situation analysis is very important before any attempts to solve the problem because:

1. It provides an opportunity to understand the dynamics of the community.
2. It helps to clarify social, economic, cultural and political conditions.
3. It provides an initial opportunity for people's participation in all project activities.
4. It enables the definition of community problems and solutions.
5. It provides information needed to determine objectives, plan and implement.

4.17 Setting Goals and Objectives

Goal setting is to ask the question, "Where do we want to go?" (What do we want?).

Before any attempts to implement a project, the planners, implementers and beneficiaries should set up goals and objectives.

A goal is a general statement of what should be done to solve a problem. It defines broadly, what is expected out of a project. A goal emerges from the problem that needs to be addressed and signals the final destination of a project. Objectives are finite sub-sets of a goal and should be specific, in order to be achievable. The objectives should be "SMART."

To achieve the objectives of a project, it is essential to assess the resources available within the community and from external sources. The goals and objectives provide the basis for monitoring and evaluating a project. They are the yardsticks upon which project success or failure is measured.

4.18 Generating Structures and Strategies

This aspect is to ask the question, "How do we get there?" (How do we get what we want with what we have?).

The planners and implementers should decide on how they are going to implement a project, which is the strategy. Agreeing on the strategy involves determining all items (inputs) that are needed to carry out the project.

Generating the structures and strategies involves:

1. Discussing and agreeing on the activities to be undertaken during implementation.
2. Defining the different roles and role players, inside and outside the community.
3. Defining and distributing costs and materials necessary to implement the project.

After establishing the appropriateness of the decisions, the executive should discuss and agree with all role players on how the project will be implemented. This is called designing a work plan. A work plan is a description of the necessary activities set out in different stages, with rough indication of the timing.

The work plan is a guide to project implementation and a basis for project monitoring. It helps to:

1. Finish the project in time.
2. Do the right things in the right order.
3. Identify responsible person/ team for all activities.
4. Determine when to start project implementation.

4.19 Implementation

Monitoring implantation is to ask the question "What happens when we do?"

Implementation is the stage where all the planned activities are put into action. Before the implementation of a project, the implementers should identify their strength and weakness, opportunities and threats (SWOT).

The strength and opportunities are positive forces that should be exploited to efficiently implement a project. The weakness and threats are obstacles that can hamper project implantation. The implementers should ensure that they devise a means of overcoming them.

The monitoring activities should appear on the work plan and should involve all stake holders. If activities are not going on well, arrangements should be made to identify the problems so that they can be corrected.

4.20 What is Project Evaluation?

Project Evaluation is a step-by-step process of collecting, recording and organising information about project results. The project results may include short-term outputs (immediate results of activities, or project deliverables), and long-term project outputs (changes in behaviour, practice or policy resulting from the project).

Need for conducting an evaluation are:

1. Response to demands of the project for accountability.
2. Demonstration of effective, efficient and equitable use of financial and other resources.
3. Recognition of actual changes and progress made.
4. Identification of success factors, need for improvement where expected outcomes are unrealistic.
5. Validation for project staff and partners when desired outcomes are achieved.

4.21 Why is Project Evaluation Important?

Evaluating project results is helpful in providing answers to questions like;

1. What progress has been made?
2. Weather the desired outcome is achieved? If not Why?

3. Are there ways that project activities can be refined to achieve better outcomes?
4. Do the project results justify the project inputs?

4.22 What are the Challenges in Monitoring and Evaluation?

1. Getting the commitment to do it.
2. Establishing base lines at the beginning of the project; Identifying realistic, quantitative and qualitative indicators.
3. Finding the time to do it and sticking to it.
4. Getting feedback from your stakeholers.
5. Reporting back to your stakeholders.

Questions:

Remember:

1. What is project planning?
2. What is the necessity of project planning?
3. List the steps in project planning.
4. What are functions of project planning?
5. Draw the project planning structure.
6. List the principle of project policies.
7. List the project planning tools.
8. What is network diagram?
9. What is project scheduling?
10. List the efforts to be taken for monitoring the time aspects of the project.
11. What is project monitoring?
12. List the information necessary to understand the community.
13. What is goal?
14. What is Project Evaluation?
15. Why is Project Evaluation Important?
16. What are the Challenges in Monitoring and Evaluation?

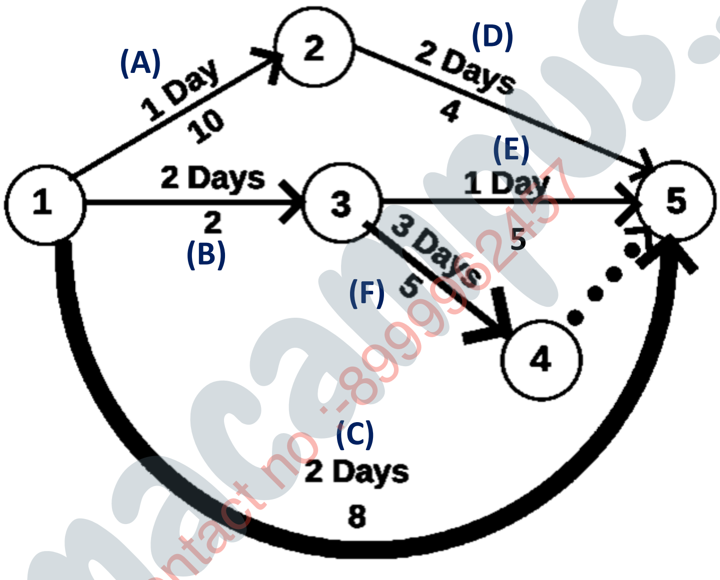
Understanding:

1. Explain the types of project plan.
2. Explain the term SMART with respect to project objectives.
3. Explain the project planning tools.
4. Explain the network diagram with example.
5. Explain Early start and Late start schedule.
6. Draw the Gantt chart for the given project.

PROJECT DETAILS

JOBS	START DAY	DURATION	MAN POWER
J-1	0	5	7
J-2	2	3	3
J-3	4	6	9
J-4	8	4	2
J-5	11	4	4

7. Design a project schedule for the given network diagram using Gantt chart. **Consider only 10 men are available for the project.**



8. Prepare an early start schedule network diagram for the given project.

ACTIVITY	PREDECESSOR	DURATION
A	--	6
B	A	6
C	A	11
D	B, C	16

9. Design a project plan to match the release of funds. The government has decided to release ₹ 1,66,00,000 required for the project in the following manner. ₹ 78,00,000 in the first year ₹ 70,00,000 in the second year, and ₹ 18,00,000 in the third year. It has also stipulated that the unspent amount would lapse and hence cannot be carried forward.

Cost estimates:

Activity	Duration (months)	Cost / month	Cost / activity
A (1-2)	10	3,00,000	30,00,000
B (1-3)	14	4,00,000	56,00,000
C (2-4)	3	8,00,000	24,00,000
D (3-4)	10	3,00,000	30,00,000
E (2-5)	10	1,00,000	10,00,000
F (4-5)	2	8,00,000	16,00,000
		Total	₹ 1,66,00,000

10. Explain the importance of situation analysis.
11. Explain the steps involved in generating the structures and strategies of a project.
12. Explain project implementation.